Anish Nethi

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Education

The University of Texas at Austin — *Aug 2024 - Present*
Master of Science in Computer Science — CGPA: 4.0/4.0
Focus Areas: AI, ML, Computer Vision, Natural Language Processing, Data Science
Coursework: Scientific Computation in ML/DL, Distributed Computing, Database Systems

Mahindra University, École Centrale School of Engineering — *Aug 2020 - Jun 2024*
Bachelor of Technology in Computer Science and Engineering — CGPA: 8.84/10.0, Rank: 5/214
Related Coursework: DL, ML with Python, Digital Image Processing, Mathematical Modeling in Image Processing, Optimization Techniques for AI, Probability and Statistics, Linear Algebra, Data Structures and Algorithms

Technical Skills

Languages: Python, C/C++, MATLAB, HTML, CSS, SQL, JavaScript
Tools & Frameworks: Jupyter, Visual Studio, Tensorflow, MySQL, SQLite, PostgreSQL, Microsoft Azure, OpenCV, PyTorch, Scikit-Learn, NumPy, Matplotlib, Pandas, Hadoop, Apache Spark, Hadoop MapReduce, Git version control, Ollama, Figma, & Latex

Experience

Indian Institute of Technology Guwahati, Research Assistant — *Jul 2023 - Jul 2024*
- Enhanced the precision of eye tracking in VR setups by 15% by leveraging data analytics techniques to differentiate saccades and fixations, generating accurate and natural scanpaths in the space-time domain

Damage in Materials and Structures Lab - Arizona State University, Research Intern — *May 2023 - Jun 2024*
- Achieved a 20% improvement in road marking classification accuracy by implementing Deep Learning and Machine Learning techniques, incorporating textural elements through pixel clustering for precise analysis

AlgoFnO, Machine Learning Intern — *Dec 2022 - Mar 2023*
- Reduced dataset size by 96% using PCA and t-SNE on TSFresh engineered features, transforming 1.2M rows into 41K rows with 50% faster processing, enhancing real-time insights with only 2% accuracy reduction

Mahindra University, Research Assistant — *Oct 2022 - Oct 2023*
- Transformed and featurized time-series forecasting problem to convert it into a regression problem for predicting volcano eruption times. Implemented various regression algorithms, improving prediction accuracy from 58% to 84%

Hewlett Packard Enterprise, Deep Learning and Cloud Solutions Intern — *Jun 2022 - Jul 2022*
- Leveraged cloud technologies to annotate and label images in Machine Learning Studio and built and deployed machine and deep learning models in a production environment on a serverless application

National University of Singapore, Academic Intern — *Jun 2022 - Jul 2022*
- Led an end-to-end BMI classification project using CNNs, incorporating ETL processes for data collection through web-scraping APIs. Enhanced data preprocessing with techniques like data augmentation, instance segmentation, and denoising, resulting in a 15% boost in model accuracy

Publications

BMI Prediction for Full Body Images using PoseNet Embeddings — *ICETCI 2023*
Detecting DeepFakes: A Deep Convolutional Neural Network Approach with Depth Wise Separable Convolutions — *ICETCI 2023*
Multiplicative Gaussian Noise Removal using Partial Differential Equations and Activation Functions: A Robust and Stable Approach — *ICACS 2023*
Cohesive Group Emotion Recognition Using Deep Learning — *SNPD 2023*

Projects

Predicting Naturalistic Eye Scanpaths from Saliency Maps — *Jul 2023 - Jul 2024*
A solution for predicting the natural path an eye would trace when viewing a scene, simulating a person's gaze
- Achieved over 83% accuracy in replicating natural eye movement patterns as measured by model performance by generating a high-frequency (90 Hz) scanpath dataset using saliency maps and time-based coordinates and developing mathematical models that capture saccades and fixations

Data Cleaning using LLMs — *Sep 2024 - Dec 2024*
A pipeline using Intel's Neural-Chat LLM for automated error detection and correction in structured datasets
- Achieved an 89.6% error correction accuracy on structured datasets by systematically integrating metadata and refining LLM prompt engineering for improved anomaly detection and correction

BMI Prediction for Full Body Images using PoseNet Embeddings — *Jun 2022 - Jul 2022*
A solution that uses ML and DL to classify a person on BMI based solely on full-body images
- Achieved 96% accuracy in BMI prediction by web-scraping full-body images, extracting features using a pre-trained PoseNet model, and applying ML algorithms, outperforming previous DL-based methods